

A02 Self-Reflection on the P+x Technique

Before implementing the code and seeing the outputs, intuitively, I expected that a smaller “x” value (which the higher rank in probabilities) would produce more reasonable sentences, while a larger value “x” is, the less sense the poem would make. The results generally supported my idea. With P+5 and P+7, GPT-2 selected tokens that were still fairly likely in the context. Although these choices were not the right ones. P+7 was slightly stranger than P+5, but the modified lines could usually still be read as sentences.

P+95 produced a much more fragmented and unpredictable result. Some replacements were even subword tokens such as “t,” “fl,” or “pl” instead of complete words, because a token can be a whole word, part of a word, punctuation, or even a space followed by some letters. By choosing the token ranked ninety-fifth, the program avoided the model’s most likely choices and selected something much less suitable.

The current program only replaces the final word of each line. To implement a P+7 technique that replaces all nouns, I would first use a part-of-speech tagging tool, such as spaCy, to identify every noun in the poem. For each noun, I would give the model the words that appear before it and calculate the next-token probabilities. I would then choose the seventh-highest alternative that can function as a noun, instead of automatically accepting any token in seventh place. I would also filter out punctuation, incomplete subwords, and the original noun. Finally, I would preserve capitalization and plural forms when inserting the replacements. This version would change the poem more consistently while keeping its basic sentence structure readable.